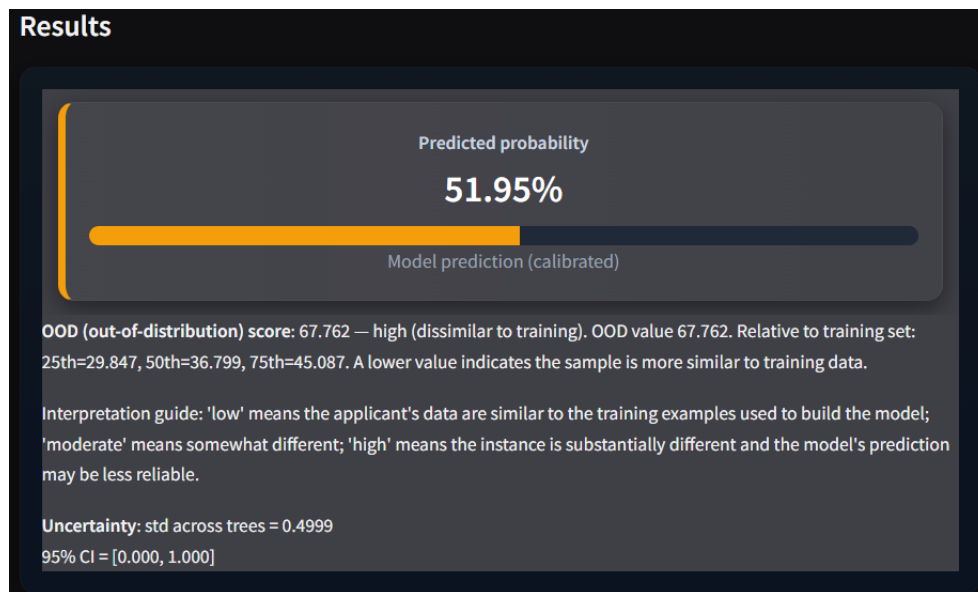


UPDATED 'AIDOPTION' PLATFORM

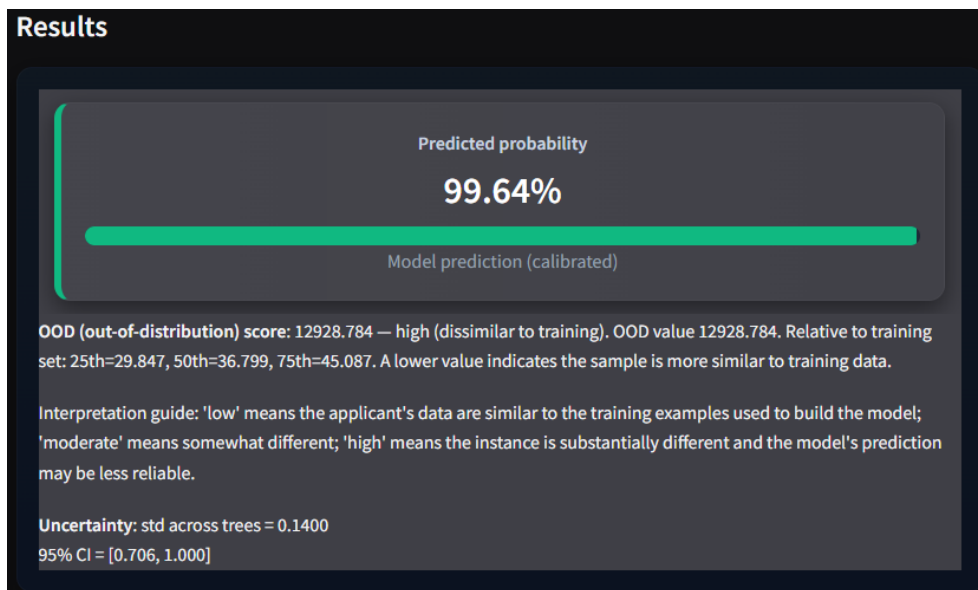
USER EXPERIENCE DESIGN PROJECT WORK

Diego Rossi

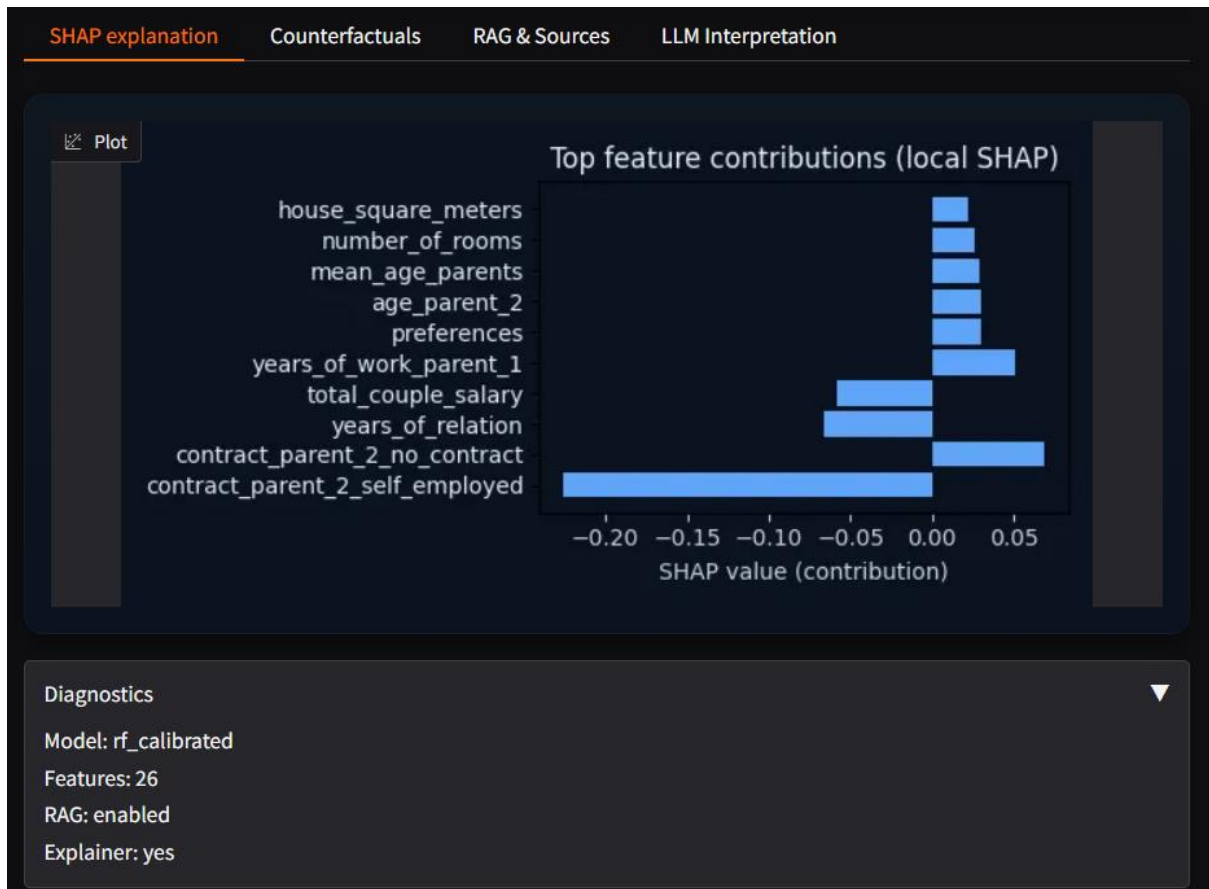
In the updated version for the AIdoption platform, results are evidenced in a clearer way for users to understand. OOD score is reported, as well as uncertainty (not very informative in this specific case).



Here I reported another case in which the uncertainty and the associated range assume more significance (even if the OOD score signals it is highly out-of-distribution):



Moreover, a graphic representation for SHAP explanation is provided. Diagnostics for the model are presented as well to the users.



A counterfactual analysis is available for the couple to understand how different choices/situations would've impacted on their probability score. A disclaimer is present as well as not every variable can be arbitrarily determined by the users.

feature	current	suggested	new_prob	delta_prob
house_square_meters	66.8	147.65641025641025	0.82	0.32999999999999996

Methodological disclaimer on counterfactual suggestions

The counterfactual modifications shown above are generated by an algorithmic local sensitivity analysis that identifies input variations which increase the model's predicted probability.

These suggested changes:

- are derived solely from the statistical relationships learned by the trained model and represent local sensitivity, not causal interventions;
- do not automatically account for practical, legal, financial, medical, or ethical feasibility constraints;
- may involve variables that are costly, difficult, or impossible to change in practice (e.g., demographic attributes or structural housing characteristics);
- do not imply that the proposed change would cause the outcome in reality.

Interpret counterfactuals as analytic aids for model transparency and hypothesis generation, not as prescriptive recommendations. Assess applicability by integrating feasibility checks, domain expertise, and causal validation.

A contextualized RAG justification, based on legal information coming from official Italian channels (Presidenza del Consiglio dei Ministri – “NORMATTIVA: il portale della legge vigente”) and enriched by a LLM is presented to the couple to have a better understanding on the mandatory constraints that they have to respect and also on how the validation process works.

RAG justification (contextualized):

Based on the provided documents, the key factors and regulations that influence adoption approval, particularly in relation to housing size and square footage requirements, can be identified as follows:

The couple's stable financial situation and moderate income are considered positive factors in their adoption prospects (Source: [DOC 2] Affidamento preadottivo). However, the documents do not explicitly mention housing size and square footage requirements as factors in adoption approval. Instead, they focus on the couple's ability to provide a stable and suitable environment for the child.

According to [DOC 2] Affidamento preadottivo, the Tribunale verifies the adoptions' requirements, including the couple's economic situation, to determine their suitability as adoptive parents. This implies that the couple's financial stability and moderate income are taken into account during the adoption process.

The documents do not provide specific information on housing size and square footage requirements for adoption approval. However, [DOC 3] Affidamento del minore mentions that the child is preferably placed in a family or a person who can provide for their maintenance, education, and instruction. This suggests that the couple's ability to provide a suitable living environment is a crucial factor in adoption approval.

In conclusion, while the documents do not explicitly mention housing size and square footage requirements, they emphasize the importance of the couple's financial stability, moderate income, and ability to provide a suitable environment for the child in the adoption process.

Sources:

- Dichiarazione di adottabilità (Presidenza del Consiglio dei Ministri - dichiarazione di adottabilità)
- Affidamento preadottivo (Presidenza del Consiglio dei Ministri - affidamento preadottivo)
- Affidamento del minore (Presidenza del Consiglio dei Ministri - affidamento del minore)

In the end, the LLM that was already provided in the first version of the platform is improved by making it possible for the users to decide the level of expertise for the generated analysis between 'basic', 'intermediate' and 'advanced'.

Expertise level (controls explanation complexity)

☐ basic ☐ intermediate ☒ advanced

In the following case, advanced level was chosen:

Based on the provided information, I'll analyze the counterfactual suggestion and provide practical explanations, feasibility notes, and prioritized recommendations.

Counterfactual Suggestion:

- **house_square_meters:** 66.8 -> 147.65641025641025 (new_prob 82.00%, Δ 33.00%)

Practical Explanation:

Increasing the square footage of the home could positively impact the adoption probability by providing a more spacious living environment for a child. This could be perceived as a more stable and comfortable setting for a child, which may be attractive to the adoption agency.

Realistic Feasibility Notes:

- The couple's current home is 66.8 square meters, which is a relatively modest size. Increasing the square footage to 147.65641025641025 square meters would require a significant renovation or expansion of their current home.
- The feasibility of this suggestion depends on various factors, including the availability of space, local building regulations, and the couple's financial resources.
- A more realistic approach might be to focus on improving the current living space or exploring alternative housing options that meet the couple's needs.

Prioritized Recommendations:

- 1. Assess the feasibility of home expansion or renovation:** The couple should consult with local authorities and architects to determine the feasibility of expanding their current home or renovating it to meet the desired square footage.
- 2. Explore alternative housing options:** If expanding or renovating the current home is not feasible, the couple should consider exploring alternative housing options that meet their needs and provide a more spacious living environment for a child.
- 3. Focus on improving the current living space:** Instead of focusing on increasing the square footage, the couple could focus on improving the current living space by decluttering, organizing, and creating a more comfortable and welcoming environment for a child.